

MATH 3150 — Notes

1 Subsequences

Definition 1.1 (Subsequence). Let $(x_n)_{n=1}^{\infty}$ be a sequence of real numbers, and let $(n_k)_{k=1}^{\infty}$ be a sequence of natural numbers that is **strictly increasing**:

$$n_1 < n_2 < n_3 < \cdots.$$

The sequence $(x_{n_k})_{k=1}^{\infty} = (x_{n_1}, x_{n_2}, x_{n_3}, \dots)$ is called a **subsequence** of (x_n) .

Equivalently, a subsequence is $x \circ \varphi$, where $x : \mathbb{N} \rightarrow \mathbb{R}$ is the sequence and $\varphi : \mathbb{N} \rightarrow \mathbb{N}$, $\varphi(k) = n_k$, is strictly increasing.

Remark 1.2.

- The *index* of a subsequence is k , not n . The k -th term of the subsequence is x_{n_k} .
- A subsequence keeps terms **in their original order** and never repeats an index. You may skip terms, but you may not reorder them or reuse them.
- Every sequence is a subsequence of itself (take $n_k = k$).

Lemma 1.3. If (n_k) is a strictly increasing sequence in $\mathbb{N}$, then $n_k \geq k$ for all $k \in \mathbb{N}$.

Proof. Induction on k . For $k = 1$: $n_1 \in \mathbb{N}$, so $n_1 \geq 1$. If $n_k \geq k$, then $n_{k+1} > n_k \geq k$, and since n_{k+1} is an integer, $n_{k+1} \geq k + 1$. $\square$

Example 1.4. Let $x_n = (-1)^n$. Then

- $n_k = 2k$ gives the subsequence $x_{2k} = 1, 1, 1, \dots$;
- $n_k = 2k - 1$ gives the subsequence $x_{2k-1} = -1, -1, -1, \dots$.

Example 1.5. Let $x_n = \frac{1}{n}$. With $n_k = k^2$, the subsequence is $x_{k^2} = \frac{1}{k^2}$, i.e. $1, \frac{1}{4}, \frac{1}{9}, \dots$.

Example 1.6 (Non-examples). For $x_n = \frac{1}{n}$:

- $(\frac{1}{2}, 1, \frac{1}{4}, \frac{1}{3}, \dots)$ is *not* a subsequence, because the terms are out of order ($n_1 = 2 > n_2 = 1$).
- $(1, 1, \frac{1}{2}, \frac{1}{2}, \dots)$ is *not* a subsequence, because indices repeat ($n_1 = n_2 = 1$).

Theorem 1.7. If $x_n \rightarrow L$, then every subsequence (x_{n_k}) also converges to L .

Proof. Let $\varepsilon > 0$. Since $x_n \rightarrow L$, there is N with $|x_n - L| < \varepsilon$ for all $n \geq N$. If $k \geq N$, then $n_k \geq k \geq N$ by Lemma 1.3, so $|x_{n_k} - L| < \varepsilon$. $\square$

Consequence. If a sequence has two subsequences with different limits, it diverges. For example, $(-1)^n$ diverges because $x_{2k} \rightarrow 1$ and $x_{2k-1} \rightarrow -1$.

1.1 Monotone subsequences

Theorem 1.8 (Monotone Subsequence Theorem). Every sequence (x_n) of real numbers has a monotone subsequence.

Proof design

Step 1: the key definition.

Definition 1.9 (Peak). An index $m \in \mathbb{N}$ is a **peak** of (x_n) if

$$x_m \geq x_n \quad \text{for all } n > m.$$

Intuition: standing at position m and looking right, nothing ahead is higher than x_m .

Step 2: split into cases. Either (x_n) has infinitely many peaks or only finitely many. These cases are exhaustive, so handling both proves the theorem.

Case 1: infinitely many peaks $\Rightarrow$ decreasing subsequence. List the peaks in increasing order, $m_1 < m_2 < m_3 < \dots$, and take $n_k = m_k$.

- *Is it a subsequence?* Check the definition: the indices must be strictly increasing.
- *Why is it monotone?* Compare x_{m_k} and $x_{m_{k+1}}$. What does “ m_k is a peak” say about every term after position m_k ? Is m_{k+1} after m_k ?

Case 2: finitely many peaks $\Rightarrow$ increasing subsequence. Choose N larger than every peak, so no index $\geq N$ is a peak.

- Start with $n_1 = N$.
- n_1 is *not* a peak. Negate the definition carefully: this gives some $n_2 > n_1$ satisfying what inequality?
- $n_2 > n_1 \geq N$, so n_2 is not a peak either. Repeat.

This is a **recursive construction**: from n_k (not a peak) we obtain $n_{k+1} > n_k$ with $x_{n_{k+1}} > x_{n_k}$. Check:

1. the indices are strictly increasing (where does that come from?);
2. the terms are increasing (strictly or non-strictly?).

Step 3: things to check in the write-up.

- **Negating “peak”**: the negation of “ $\forall n > m, x_m \geq x_n$ ” is “ $\exists n > m$ with $x_n > x_m$.” Case 2 depends on this.
- $\geq$ **vs.** $>$: if a peak is defined with $>$ instead, Case 1 gives a *strictly* decreasing subsequence and Case 2 only a *non-decreasing* one. Either works if used consistently.
- **Case 2 is an induction**: say explicitly that n_{k+1} is defined from n_k .

Why it matters. Combined with the Monotone Convergence Theorem, this gives **Bolzano–Weierstrass**: a bounded sequence has a monotone subsequence (Theorem 1.8), which is bounded, hence converges.

My proof

1.2 Bolzano–Weierstrass

Theorem 1.10 (Bolzano–Weierstrass). Every bounded sequence of real numbers has a convergent subsequence.

Proof setup

Hypothesis and goal.

- **Given:** (x_n) is bounded, i.e. there is $M > 0$ with $|x_n| \leq M$ for all $n \in \mathbb{N}$.
- **Want:** a strictly increasing sequence of indices (n_k) such that (x_{n_k}) converges.

Tools available.

- **Theorem 1.8:** every sequence has a monotone subsequence.
- **Monotone Convergence Theorem (MCT):** a sequence that is monotone *and* bounded converges.

The chain of steps. The proof links these two results. Fill in the justification for each step.

1. **Get a monotone subsequence.** Apply Theorem 1.8 to (x_n) to obtain a monotone subsequence (x_{n_k}) . *Did this step use boundedness?*
2. **Show the subsequence is bounded.** Why does $|x_{n_k}| \leq M$ hold for every k ? *Hint: is x_{n_k} a term of the original sequence?*
3. **Apply MCT.** Steps 1 and 2 verify both hypotheses of MCT for (x_{n_k}) . Conclude.
4. **Match the goal.** (x_{n_k}) is a subsequence of (x_n) and it converges, which is exactly what the theorem asks for.

Things to check.

- **Where is boundedness used?** Only in Step 2. Without it, the statement fails: $x_n = n$ has monotone subsequences, but none of them converge.
- **The limit need not be unique across subsequences.** For $(-1)^n$, B–W guarantees *some* convergent subsequence, but different subsequences converge to 1 and -1 .
- **Where does the limit live?** If $|x_n| \leq M$, what can you say about the limit L of (x_{n_k}) ? (Is $|L| \leq M$?)

Alternative approach (bisection). Another standard proof avoids Theorem 1.8. Start with $[-M, M]$, split it in half, and keep a half that contains x_n for *infinitely many* n . Repeat to get nested intervals whose lengths go to 0, and pick one index from each interval, choosing indices that increase. Worth comparing the two proofs once the first one is done.

My proof

1.3 Notes

We need to use definitnos to set up our proofs.